	AzureSky	
	Determination of carbon in ash	Document no: AS016IW
	Compiled by:	**Sifiso Nyambi - Laboratory Manager**
	Approved by:	**Nonhlanhla Msibi - Operations Manager**

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1. PRE REQUISITES

note: Only personnel who are declared competent are to perform laboratory activities. personnel not declared are to strictly work under supervision of a lab manager/supervisor or competent personnel.

- Ensure that all wet coarse and fly ash samples are dried in an oven 105oC for 1 hour before testing.
- Ensure that coarse ash samples are ground down to be fine.
- Personnel must always wear a mask during sample handling.

2. APPARATUS AND EQUIPMENT REQUIRED

- Analytical balance
- ISO furnace
- Pestle and mortar
- Watch glass and spatula
- Heat resistant crucibles
- Pair of tongs with hand shield

3. PROCEDURE (FURNACE METHOD)

3.1. Sample preparation

- Upon receipt of the samples, transfer wet coarse ash into a tray and dry in an oven set to 105°C.
- After drying and cooling, mix thoroughly to obtain a representative sample.
- Grind the coarse ash using pestle and mortar until it is fine enough to pass through a 75µm mesh

3.2. Analysis method

- Weigh 1g of sample in duplicate into clean, dry crucible, and spread the material evenly on the surface of the crucible.
- Use a pair of tongs and carefully place the samples into the ISO Ash furnace.
- Press RUN button on the furnace controls to initiate the test cycle.
- Allow the ISO furnace to ramp and reach 815°C until END appears on the display panel.
- Cool for 10 minutes on the bench, and 10 minutes in the desiccator.
- Weigh the final mass of crucible+contents.

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4. Calculation

%Carbon = (100- $\frac{M3-M1}{M2-M1}$) $\boldsymbol{X\ 100}$

5. Revision table

Revision no	Content changed	Reason for revision	Date changed

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